

Object Detection Using Faster R-CNN on PASCAL VOC 2012

1 Introduction

Object detection is a fundamental problem in computer vision that involves identifying and localizing objects of interest within an image. Unlike image classification, object detection requires predicting both the **class label** and the **bounding box coordinates** of each object present in the scene.

In this project, I implement and train a **Faster R-CNN** based object detection model using the **PASCAL VOC 2012** dataset. The goal is to detect multiple object categories in natural images using a deep learning-based region proposal and classification framework.

2 Dataset Description

2.1 Dataset Used

- **PASCAL VOC 2012**
- Total images (train split): ~ 5717
- Images contain objects from 20 categories

2.2 Selected Classes

For this task, the dataset was filtered to include the following **five object classes**:

Class Name	Label ID
Person	1
Car	2
Dog	3
Cat	4
Bottle	5

Images that did not contain at least one object belonging to the selected classes were **discarded** during dataset loading.

3 Data Preprocessing & Augmentation

3.1 Annotation Parsing

XML annotations were parsed to extract:

- Bounding box coordinates (`xmin`, `ymin`, `xmax`, `ymax`)
- Corresponding class labels

Each sample returns:

- Image tensor
- Target dictionary containing `boxes`, `labels`, `area`, and `iscrowd`

3.2 Data Augmentation

To improve generalization and robustness, **Albumentations** was used for bounding-box-safe augmentation:

Training augmentations:

- Horizontal flip
- Random brightness & contrast
- Shift, scale, and rotation
- Normalization
- Conversion to PyTorch tensors

Validation augmentations:

- Image resizing
- Normalization
- Tensor conversion

All augmentations preserved bounding-box consistency.

4 Model Architecture

4.1 Faster R-CNN Overview

Faster R-CNN is a two-stage object detection framework consisting of:

1. **Backbone CNN** for feature extraction
2. **Region Proposal Network (RPN)** for candidate object regions
3. **ROI Align** for fixed-size feature extraction
4. **Detection Head** for classification and bounding box regression

4.2 Backbone Network

- **ResNet-50 with Feature Pyramid Network (FPN)**
- Pretrained on **ImageNet**
- Provides multi-scale feature maps for better detection of objects of varying sizes

4.3 Output Classes

- Total classes = **6**
 - 5 object classes + 1 background class

5 Training Setup

5.1 Hyperparameters

- Optimizer: **Stochastic Gradient Descent (SGD)**
- Learning rate: **0.0005**
- Momentum: **0.9**
- Weight decay: **0.0005**
- Batch size: **4**
- Epochs: **5**
- Learning rate scheduler: **StepLR**

5.2 Training Strategy

- Training performed on **Google Colab GPU**
- Loss values monitored for NaNs
- Invalid samples safely skipped
- Both **training loss** and **validation loss** tracked per epoch

6 Evaluation Metrics

6.1 Mean Average Precision (mAP)

Model performance was evaluated using **Mean Average Precision**, a standard metric in object detection.

- **mAP@0.5**: IoU threshold = 0.5
- **mAP@0.5:0.95**: Average over IoU thresholds from 0.5 to 0.95

These metrics were computed using **TorchMetrics**.

7 Results

7.1 Loss Curves

- Training loss showed a consistent decrease across epochs
- Validation loss remained stable, indicating no severe overfitting

7.2 Quantitative Results

Metric	Value
mAP@0.5	0.55
mAP@0.5:0.95	0.24

These values indicate reasonable detection performance given limited training epochs and dataset size.

7.3 Qualitative Results

- Model successfully detected all the classes in most cases
- Bounding boxes were accurately localized in most cases
- Confidence scores aligned with visual correctness

Sample detections were visualized and screen-recorded as part of the submission.

8 Observations & Challenges

- **Class imbalance** significantly affected minority classes (e.g., dog and cat)
- Limited compute resources restricted the number of training epochs
- Longer training and balanced data could further improve performance
- Lower mAP@0.5:0.95 is expected due to stricter localization requirements

9 Conclusion

In this project, a Faster R-CNN object detection model was successfully implemented and trained on a subset of the PASCAL VOC 2012 dataset. The model demonstrated effective localization and classification of multiple object categories. Despite limited training epochs and dataset imbalance, the model achieved reasonable mAP scores and produced visually accurate detections. This work highlights the effectiveness of region-based convolutional neural networks for object detection tasks and provides a strong foundation for further improvements.

10 Future Work

- Increase training epochs with stronger hardware
- Use class-balanced sampling
- Fine-tune anchor sizes
- Experiment with alternative backbones
- Apply transfer learning from COCO-trained models